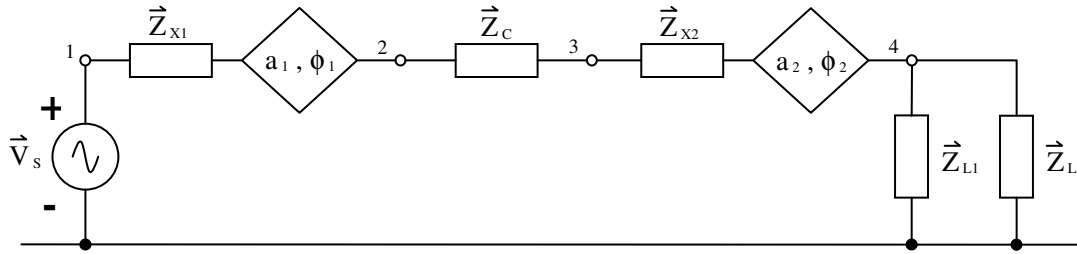


Problem #1

The equivalent circuit for a 3 ϕ power system and the numeric values for its circuit parameters are given below.



$$\bar{V}_S = \left(\frac{485.125}{\sqrt{3}} \right) \angle 0^\circ \text{ V}, \quad \bar{Z}_{L1} = 10 \angle -35^\circ \Omega, \quad \bar{Z}_{L2} = \frac{1}{3} (33 \angle 70^\circ \Omega), \quad \bar{Z}_C = 25.35 \angle 77^\circ \Omega$$

$$a_{x1} = \frac{480}{8000}, \quad \bar{Z}_{x1} = \left(\frac{0.081 + j0.097}{3} \right) + a_{x1}^2 (0.01 + j0.1) = 0.04242 \angle 50.411^\circ \Omega, \quad \phi_{x1} = 30^\circ$$

$$a_{x2} = \frac{8000}{480}, \quad \bar{Z}_{x2} = \left(\frac{0.05 + j0.4}{3} \right) + a_{x2}^2 \left(\frac{0.092 + j0.053}{3} \right) = 9.91254 \angle 30.565^\circ \Omega, \quad \phi_{x2} = 60^\circ$$

A sequence of eight circuit analysis equations for doing certain specific calculations in this power system problem are given below. Using your scientific calculator, first enter the circuit parameters defined above and store them as variables in your calculator. Then using these stored variables, enter the analysis equations into your calculator one at a time, and store the results of all eight analysis equations as variables in your calculator.

$$\bar{Z}_{eq3} = \bar{Z}_{x2} + a_{x2}^2 (\bar{Z}_{L1}^{-1} + \bar{Z}_{L2}^{-1})^{-1}$$

$$\bar{Z}_{eq2} = \bar{Z}_C + \bar{Z}_{eq3}$$

$$\bar{I}_S = \bar{V}_S (\bar{Z}_{x1} + a_{x1}^2 \bar{Z}_{eq2})^{-1}, \quad \bar{S}_S = \bar{V}_S \bar{I}_S^*$$

$$\bar{I}_{PC} = (a_{x1} \angle \phi_{x1}) \bar{I}_S$$

$$\bar{V}_2 = \bar{Z}_{eq2} \bar{I}_{PC}, \quad \bar{V}_3 = \bar{Z}_{eq3} \bar{I}_{PC}, \quad \bar{V}_4 = (a_{x2}^{-1} \angle \phi_{x2}) (\bar{V}_3 - \bar{Z}_{x2} \bar{I}_{PC})$$

The sequence of equations for calculating several desired final results from the equivalent circuit analysis results are given below. Use your calculator to enter these equations one at a time, and verify your numeric results with what is shown below.

$$P_{3\phi S} = 3 \cdot \text{Re}\{\bar{S}_S\} = 26,168.352 \text{ W}$$

$$Q_{3\phi S} = 3 \cdot \text{Im}\{\bar{S}_S\} = 6,873.940 \text{ VAR}$$

$$V_{LL2} = \sqrt{3} \cdot |\bar{V}_2| = 8053.423 \text{ V}$$

$$I_{L2} = |\bar{I}_{PC}| = 1.932 \text{ A}$$

$$\theta_2 = \angle \bar{V}_2 - \angle \bar{I}_{PC} = 14.554^\circ$$

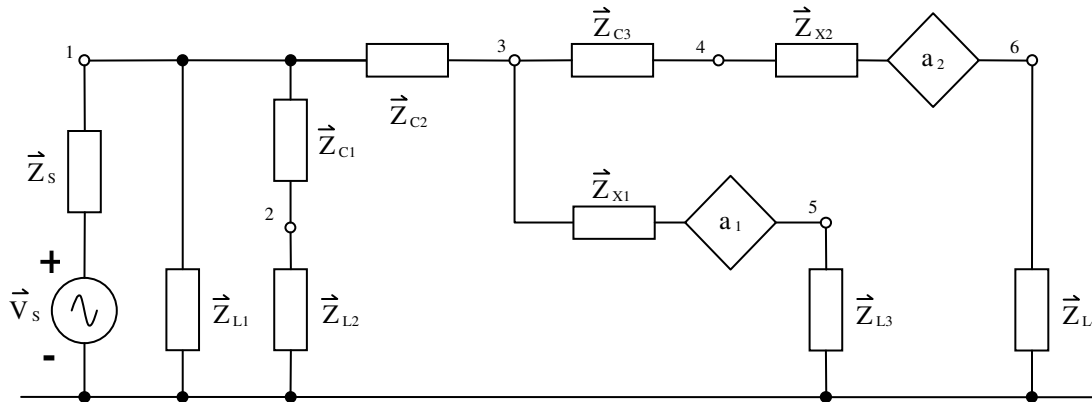
$$V_{LL3} = \sqrt{3} \cdot |\bar{V}_3| = 8014.535 \text{ V}$$

$$I_{L3} = |\bar{I}_{PC}| = 1.932 \text{ A}$$

$$\theta_3 = \angle \bar{V}_3 - \angle \bar{I}_{PC} = 14.017^\circ$$

Problem #2

The equivalent circuit for a 1 ϕ power system and the numeric values for its circuit parameters are given below.



$$\vec{V}_s = 962.35 \angle 0^\circ \text{ V}, \quad \vec{Z}_s = 0.001 \angle 86.75^\circ \Omega$$

$$\vec{Z}_{L1} = 50 \angle 20^\circ \Omega, \quad \vec{Z}_{L2} = 80 \angle 65^\circ \Omega, \quad \vec{Z}_{L3} = 10 \angle 70^\circ \Omega, \quad \vec{Z}_{L4} = 5 \angle 30^\circ \Omega$$

$$\vec{Z}_C = 0.000058 \angle 42.53^\circ \Omega / \text{ft}, \quad \vec{Z}_{C12} = 2(1300 \text{ ft}) \vec{Z}_C, \quad \vec{Z}_{C13} = 2(350 \text{ ft}) \vec{Z}_C, \quad \vec{Z}_{C34} = 2(975 \text{ ft}) \vec{Z}_C$$

$$a_{x1} = 960/480, \quad \vec{Z}_{x1} = (0.01 + j0.12) + a_{x1}^2 (0.003 + j0.07)$$

$$a_{x2} = 960/240, \quad \vec{Z}_{x2} = (0.08 + j0.49) + a_{x2}^2 (0.002 + j0.03)$$

A sequence of circuit analysis equations for doing certain specific calculations in this power system problem are given below. Using your scientific calculator, first enter the circuit parameters defined above and store them as variables in your calculator. Then using these stored variables, enter the analysis equations into your calculator one at a time, and store the results of all analysis equations as variables in your calculator.

$$\vec{Z}_{eq4} = \vec{Z}_{x2} + a_{x2}^2 \vec{Z}_{L4}$$

$$\vec{Z}_{eq3} = \left[(\vec{Z}_{x1} + a_{x1}^2 \vec{Z}_{L3})^{-1} + (\vec{Z}_{C34} + \vec{Z}_{eq4})^{-1} \right]^{-1}$$

$$\vec{Z}_{eq1} = \left[\vec{Z}_{L1}^{-1} + (\vec{Z}_{C12} + \vec{Z}_{L2})^{-1} + (\vec{Z}_{C13} + \vec{Z}_{eq3})^{-1} \right]^{-1}$$

$$\vec{I}_s = \vec{V}_s (\vec{Z}_s + \vec{Z}_{eq1})^{-1}, \quad \vec{V}_1 = \vec{V}_s - \vec{Z}_s \vec{I}_s, \quad \vec{S}_s = \vec{V}_1 \vec{I}_s^*$$

$$\vec{I}_{12} = \vec{V}_1 (\vec{Z}_{C12} + \vec{Z}_{L2})^{-1}, \quad \vec{V}_2 = \vec{V}_1 - \vec{Z}_{C12} \vec{I}_{12}$$

$$\vec{I}_{13} = \vec{V}_1 (\vec{Z}_{C13} + \vec{Z}_{eq3})^{-1}, \quad \vec{V}_3 = \vec{V}_1 - \vec{Z}_{C13} \vec{I}_{13}$$

$$\vec{I}_{34} = \vec{V}_3 (\vec{Z}_{C34} + \vec{Z}_{eq4})^{-1}, \quad \vec{V}_4 = \vec{V}_3 - \vec{Z}_{C34} \vec{I}_{34}$$

$$\vec{I}_{x11} = \vec{V}_3 (\vec{Z}_{x1} + a_{x1}^2 \vec{Z}_{L3})^{-1}, \quad \vec{I}_{x12} = a_{x1} \vec{I}_{x11}, \quad \vec{V}_5 = \vec{Z}_{L3} \vec{I}_{x12}$$

$$\vec{I}_{x21} = \vec{V}_4 (\vec{Z}_{x2} + a_{x2}^2 \vec{Z}_{L4})^{-1}, \quad \vec{I}_{x22} = a_{x2} \vec{I}_{x21}, \quad \vec{V}_6 = \vec{Z}_{L4} \vec{I}_{x22}$$

The sequence of equations for calculating several desired final results from the equivalent circuit analysis results are given below. Use your calculator to enter these equations one at a time, and verify your numeric results with what is shown below.

$$P_s = \text{Re}\{\vec{S}_s\} = 39,940.817 \text{ W}, \quad Q_s = \text{Im}\{\vec{S}_s\} = 44,166.488 \text{ VAR}$$

$$V_1 = |\vec{V}_1| = 962.302 \text{ V}, \quad V_2 = |\vec{V}_2| = 960.628 \text{ V}, \quad V_3 = |\vec{V}_3| = 960.973 \text{ V}$$

$$V_4 = |\vec{V}_4| = 959.655 \text{ V}, \quad V_5 = |\vec{V}_5| = 475.923 \text{ V}, \quad V_6 = |\vec{V}_6| = 238.170 \text{ V}$$

Problem #3

Perform the following calculations using your scientific calculator and storing things as variables in your calculator where appropriate.

$$R = 0.1234 \, \Omega / \text{mile}, \quad G = 0, \quad GMR = 0.0355 \, \text{ft}, \quad mGMR = 0.0438 \, \text{ft}$$

$$GMD_{eq} = \sqrt[3]{(15)(12)(\sqrt{15^2 + 12^2})}$$

$$L = 3.218688 \times 10^{-4} \cdot \ln(GMD_{eq}/GMR)$$

$$C = 8.953181 \times 10^{-8} / \ln(GMD_{eq}/mGMR)$$

$$\bar{\mathcal{Z}} = R + j120\pi L$$

$$\bar{y} = G + j120\pi C$$

$$\bar{V}_S = \left(\frac{230,000}{\sqrt{3}} \right) \angle 0^\circ \text{ V}, \quad \bar{Z}_{L1} = 350 \angle 75^\circ \, \Omega, \quad \bar{Z}_{L2} = \frac{1}{3}(990 \angle -90^\circ \, \Omega), \quad \bar{Z}_{L3} = 1100 \angle 65^\circ \, \Omega$$

$$\bar{Z}_{TL1} = 50 \cdot \bar{\mathcal{Z}}$$

$$\bar{Y}_{TL1} = \frac{1}{2}(50) \cdot \bar{y}$$

$$\bar{Z}_{TL2} = 10 \cdot \bar{\mathcal{Z}}$$

$$\bar{Z}_{TL3} = 20 \cdot \bar{\mathcal{Z}}$$

$$\bar{Z}_{eq3} = \left[\bar{Z}_{L2}^{-1} + (\bar{Z}_{TL3} + \bar{Z}_{L3})^{-1} \right]^{-1}$$

$$\bar{Z}_{eq2} = \left[\bar{Y}_{TL1} + \bar{Z}_{L1}^{-1} + (\bar{Z}_{TL2} + \bar{Z}_{eq3})^{-1} \right]^{-1}$$

$$\bar{I}_Z = \bar{V}_S (\bar{Z}_{TL1} + \bar{Z}_{eq2})^{-1}$$

$$\bar{I}_S = \bar{Y}_{TL1} \bar{V}_S + \bar{I}_Z$$

$$\bar{S}_S = \bar{V}_S \cdot \bar{I}_S^*$$

$$\bar{V}_2 = \bar{V}_S - \bar{Z}_{TL1} \bar{I}_Z$$

$$\bar{V}_3 = \bar{Z}_{eq3} (\bar{Z}_{TL2} + \bar{Z}_{eq3})^{-1} \bar{V}_2$$

$$\bar{V}_4 = \bar{Z}_{L3} (\bar{Z}_{TL3} + \bar{Z}_{L3})^{-1} \bar{V}_3$$

$$P_{S3\phi} = 3 \cdot \text{Re}\{\bar{S}_S\} = 57.988885 \text{ MW}$$

$$Q_{S3\phi} = 3 \cdot \text{Im}\{\bar{S}_S\} = 13.592997 \text{ MVAR}$$

$$V_{L1LL} = \sqrt{3} \cdot |\bar{V}_2| = 225.2215 \text{ KV}$$

$$V_{L2LL} = \sqrt{3} \cdot |\bar{V}_3| = 228.8365 \text{ KV}$$

$$V_{L3LL} = \sqrt{3} \cdot |\bar{V}_4| = 225.8863 \text{ KV}$$